

KARNATAKA RESIDENTIAL EDUCATIONAL INSTITUTIONS SOCIETY

ICT Class 9 - Practical Lab Exam 1

Practical Lab Exam 1

Class: Class 9

Assessment Type: Practical Lab Exam 1

Total Marks: 20

Duration: 45 minutes

Topics Covered: Python Programming and Data Structures

ICT Teaching Resources Portal

General Instructions

1. All questions are compulsory.
2. Read each question carefully before answering.
3. Write neatly and legibly.
4. Marks are indicated against each question.

Lab Tasks (15 marks)

Task 1: Student Grade Manager (5 marks)

Write a Python program that:

- Stores student data as a list of dictionaries with keys: name, roll_no, subjects (dict of sub: marks)
- Implements functions for:
 - Adding a new student with marks in 5 subjects
 - Updating marks for an existing student
 - Calculating grade based on average (A: 90+, B: 75-89, C: 60-74, D: 45-59, F: below 45)
 - Finding the class topper and subject toppers
 - Displaying a formatted report card for all students
- Uses proper input validation and error handling

Task 2: Inventory Management System (5 marks)

Write a Python program that:

- Uses a dictionary to store inventory items with keys: item_id, name, quantity, price, category
- Implements a menu-driven interface with options:
 - Add new item
 - Update stock (increase/decrease)
 - Search items by name or category
 - Calculate total inventory value
 - Display items sorted by price or quantity
 - Generate a low-stock alert (quantity below 5)
- Includes a function to export inventory data to a text file

Task 3: Pattern Generator (5 marks)

Write a Python program that generates the following patterns using nested loops:

- Pattern A: Right-angled triangle of stars (5 rows)

```
*
**
***
****
*****
```

- Pattern B: Pyramid pattern (5 rows)

```
    *
   ***
  *****
 *****
*****
```

- Pattern C: Number pattern (5 rows)

```
1
2 2
3 3 3
4 4 4 4
5 5 5 5 5
```

- The program should use functions for each pattern and allow the user to choose which pattern to display
- Allow custom input for the number of rows (1-10)

Viva Voce (5 marks)

1. What is the difference between a **function definition** and a **function call**? Give an example of each. (1 mark)
2. Explain the difference between **mutable** and **immutable** data types. Give two examples of each. (1 mark)
3. How does Python handle memory when you create a **shallow copy** vs a **deep copy** of a nested list? (1 mark)
4. What is the purpose of the `__init__` method in a Python class? Can you create an object without it? (1 mark)
5. Explain the difference between **linear search** and **binary search**. When would you choose one over the other? (1 mark)